

Secured Key Generation for Biometric Encryption using Hyper-chaotic Map and DNA Sequences

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ABSTRACT

Security is the major concern in this digital era communication. A novel key generation algorithm is proposed which can be used for image encryption. The suggested work uses hyper chaotic map and DNA sequences for implementation. Beta chaotic function is used for hyper chaotic functionality and scrambling is also performed to increase chaos which will enhance key security. DNA addition and complement functions are applied to the scrambled image that increases the complexity of the system. Finally, the cryptographic key is generated using SHA256 which can be used for image encryption. Multimodal biometrics is used to enhance security and accuracy. The simulation results prove that the key generated using hyper chaotic function and DNA addition is highly sensitive and protected. Several methods are performed to check the strength of key generated and hence proved the algorithm is resistant to key sensitivity analysis, information entropy analysis and correlation co-efficient analysis.

Keywords: Cryptanalysis, Chaotic maps, Hyper-chaotic, DNA sequences, Scrambling, Multimodal, Biometrics, SHA 256.

1. Introduction

Biometrics is mainly used for the automated verification of an individual based on physiological or behavioral characteristics [5, 8, 10]. Some of the biometric traits used for authentication are Fingerprints, face, DNA, palm print, hand geometry, Iris, retinal image, typing rhythm, gait, voice etc. [11]. Communications have been revamped due to the evolution of new network technologies. Securing the digital image is a big challenge in the network to protect it against malicious threats. There are several methods to secure digital image in real time applications such as cryptographic algorithms, passcodes etc. [6] where the large size of a cryptographically generated strong key, cannot be easily counterfeited but user finds it intricate to remember [7, 12] and the passcodes can also be lacerated. Cryptographic algorithm is unable to differentiate between the legitimate user and an attacker who fraudulently acquires the passcode. Biometric authentication offers an innovative means for key security which bestow direct connection to passcode and key [9].

Biometric image have high resolution, occupies large storage space, high redundancy and strong correlations among adjacent pixels because of which it is not competent with traditional encryption methods. We need a system where the image is sensitive to any small changes made to initial conditions, pseudo randomness etc. Chaotic systems meet the requirements of the encryption system to secure the image. To increase the efficiency of the encryption system DNA is used for the further encoding. Several algorithms are developed with chaotic systems and DNA, but few challenges are to make the system resistant to several malicious threats [33].

To provide the access to secured key, user's biometric trait is captured. On successful authentication, the key is released which further enables to encrypt or decrypt the desired data [12]. An approach is to encompass selected features of biometric sample as input into the encryption /decryption procedure, which support greater robustness [14, 15]. Biometric security system can be listed as uniqueness, universality, permanence, collectability, performance, acceptability and circumvention [7]. Single trait may engulf the desired performance which necessitates multimodal biometric system [32]. Superior level of safety measures can be achieved by using multi modalities of biometric.

Biometrics is used for several applications such as ticketing for events, travel boarding card with biometric, to prescribe medicines, credit/loyalty card systems, databases for medical records, delicate financial records, payment transactions, remote authentication, admission control for restricted area, securing files, drives, e-mails etc. Biometrics are also used for to access sensitive information held by government and various specific organizations, law enforcement, smart cards, secure use of cloud verification services. The key generated using chaotic maps and DNA are verified for histogram analysis to check the distribution of data after scrambling, key space analysis to check the size of the key, key sensitivity analysis to check the strength of the key, correlation analysis to know the strength of scrambling and information entropy analysis to check the randomness of scrambling. The above mentioned analysis prove that the key generated using the proposed system is highly secure and are resistant to several attacks. Multimodal biometrics is used for key generation because of the following advantages,

- A. Accuracy
 - ◆ various fusion techniques are used to combine multiple traits
 - ◆ more accurate than unimodal biometric
- B. Increased and reliable recognition
 - ◆ multiple traits are used to increase the authenticity
- C. Enhanced security
 - if one traits fails other traits can provide identification
- D. Vulnerability
 - ◆ has the ability to avoid spoofing attacks
- E. User acceptance
 - accurate, reliable, highly secure therefore it become ubiquitous

2. Existing Chaotic Maps

Different types of chaotic maps are shown in Fig.1. The properties of existing chaotic maps are explained as follows,

- A. Logistic
 - less complex, very high efficiency system, linear and non-linear dynamic function, 1D, 2D
- B. Spatiotemporal
 - highly dynamic, balances complexity and efficiency for cryptographic purposes
- C. Hyperchaotic
 - highly random, generates more Lyapunov exponents
- D. Non-linear hyperchaotic
 - creates highly chaotic image

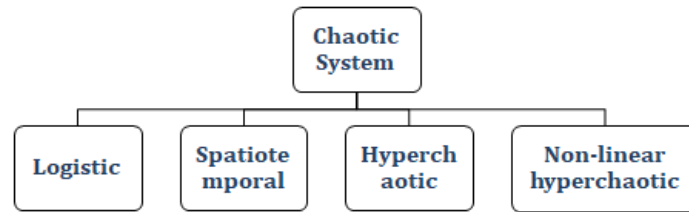


Fig. 1 - Chaotic maps.

2.1. Logistic chaotic map

Y. Zhang [1], Logistic chaotic maps have low complexity but highly efficient. Dimensionality is one of the parameter for classification of chaotic map as 1D and multi dimensionality chaotic map. High dimensional chaotic maps are suitable for image encryption. The Eqn. 1 represents 1D logistic chaotic map whose function is linear and Eqn. 2 represents 2D logistic chaotic system whose function is non-linear and dynamic,

$$X_{n+1} = f(x) = \mu X_n (1 - X_n) \quad (1)$$

where $\mu \in (0, 4)$, $X_n \in (0, 1)$ $n = 0, 1, 2, \dots$

Research results prove that 1D chaotic map behaves more chaotic when, $3.56994 < \mu \leq 4$.

$$\begin{cases} X_{i+1} = \mu_1 x_i (1 - x_i) + \gamma_1 y_i^2 \\ y_{i+1} = \mu_2 y_i (1 - y_i) + \gamma_2 (x_i^2 + y_i x_i) \end{cases} \quad (2)$$

when $2.75 < \mu_1 \leq 3.4$, $2.75 < \mu_2 \leq 3.45$, $0.15 < \gamma_1 \leq 0.21$, $0.13 < \gamma_2 \leq 0.1$ the 2D logistic system is chaotic where the method creates two sequences in the area $[0, 1]$.

2.2 Chen's hyper chaotic map

The 1D chaotic maps are combined together to form multidimensional chaotic maps which is known as hyperchaotic maps. L. zeng et.al [3], generates four chaotic sequences as shown in Eqn.3. The sequences generated are highly random with two positive Lyapunov exponents. The number of Lyapunov exponents created by the system proves chaotic nature of system. When the positive Lyapunov exponents are generated in more number, then the system will be highly chaotic.

$$\begin{cases} \dot{x} = a(y - x); \\ \dot{y} = -xz + dx + cy - q; \\ \dot{z} = xy - bz; \\ \dot{q} = x + k; \end{cases} \quad (3)$$

Assume $a = 36$, $b = 3$, $c = 28$, $d = 16$ and $-0.7 \leq k \leq 0.7$. Four Lyapunov exponents $[\lambda_1 = 1.552, \lambda_2 = 0.023, \lambda_3 = 0, \lambda_4 = -12.573]$ are obtained by solving Eqn.3 using Runge-kutta method.

2.3 5D hyper chaotic map

5D hyper chaotic map [34] is generated by adding a linear feedback control to Eqn.3 and a new five-dimensional hyperchaotic system is generated, which is shown in Eqn.4.

$$\begin{cases} \dot{x} = a(y - x); \\ \dot{y} = -xz + dx + cy - q; \\ \dot{z} = xy - bz; \\ \dot{q} = x + k; \\ \dot{r} = eyz; \end{cases} \quad (4)$$

For different values of a, b, c, d, e, x, k , the system behaves in different chaotic manner.

2.4 Beta hyper chaotic map

Rim Zahmoul et.al [4], Beta chaotic map is determined from beta function which has high chaotic behavior. The mapping is polynomial and is given in Eqn.5. The chaotic sequences generated are highly random with two positive Lyapunov exponents

$$x_{n+1} = k * \text{Beta}(x_n; x_2, p, q)$$

Where, $p = b_1 + c_1 * a$ and $q = b_2 + c_2 * a$ with b_1, c_1, b_2 and c_2 are constants. The k parameter is used to control amplitude of the beta map and a denote the bifurcation parameter. The beta is given in equation (5)

$$\text{Beta}(x_n; x_2, p, q) = \begin{cases} \left(\frac{(x-x_1)}{(x_c-x_1)} \right)^p \left(\frac{(x_2-x)}{(x_2-x_c)} \right)^q & \text{if } x \in [x_1, x_2] \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

Beta chaotic map is highly sensitive to its parameters. Beta map has strong chaotic behavior, more efficient and resistant to several attacks.

3. Existing Template Protection Techniques Using Biometric Key

Biometric template can be protected either by feature transformation or biometric cryptosystem. Some of the feature transformation techniques are salting and non-invertible transform whereas biometric cryptosystem includes key binding and key generation [13, 25].

3.1. Properties

Few Template protection properties are shown in table 1.

Table 1 – Template protection properties

Diversity	Revocability	Security	Performance
no cross matching across databases	reissue new template using the existing biometric data	computationally tough to get the original template	performance FAR, FRR should not be degraded

3.2. Techniques

Some of the techniques which are used for the template protection are shown in table 2. The methodologies used for the different techniques of template protection are discussed with its strengths and weakness [16 -18].

Table 2 – Template protection techniques

Technique	Methodology	Strengths	Weakness
Biometric Encryption	Standard cryptographic algorithms	Prevents attacker from decrypting the template	- Possible to retrieve original template - Susceptible to blended substitution, hill climbing and nearest imposter attacks
Fuzzy commitment	Cryptography and ECC	Protects secret key by storing its hash value	- Poor accuracy performance - Susceptible to ARM, hill climbing, brute force, decodability and cross matching attacks
Fuzzy Vault	Unordered set of biometric data to lock a secret key	Cannot be decoded without a biometric data	- Susceptible to ARM, brute force and collusion attacks - Vulnerable to intrusion, liaison combination and injection attacks
Shielding Function	Random secret biometric data	Cannot be decoded without key	- Short key length - Susceptible to ARM, brute force and cross matching attacks.

4. Proposed algorithm

The process of generating secured key with chaotic system and DNA sequences is depicted in Fig.3 Initially the input images are scrambled using the values generated from beta chaotic map. DNA encoding is performed for the scrambled images w

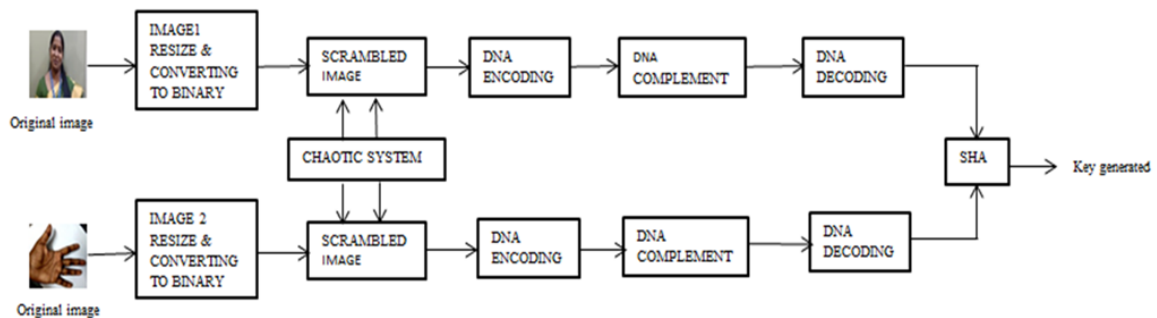


Fig 3 – Proposed key generation

To make the generated key highly secure, DNA complement rule is applied to the encoded images. To generate the fixed size key, SHA 256 is applied after decoding. The following steps illustrate the procedure for key generation,

Step 1: Input two Images and convert in to equal binary blocks. Consider an input image I is divided in to three blocks B1, B2, B3.

$$I(m,n,2)=B1(m,n)+B2(m,n)$$

Step 2: Generate chaotic sequences with initial values [x, y, z, q] of chosen chaotic systems.

Step 3: Covert the blocks in to scrambled images using chaotic sequence values [lx,ly,lz,lq]

$$B1(i,j) \leftrightarrow B1(lx(i),ly(j))$$

$$B2(i,j) \leftrightarrow B2(lx(i),lz(j))$$

$$B3(i,j) \leftrightarrow B3(ly(i),lz(j))$$

Step 4: Divide the scrambled blocks in to equal sized blocks B1b, B2b, B3b.

Step 5: Add the scrambled blocks using DNA addition.

$$B1b\{i,j\} = B1b\{i,j\} + B1(lx(i),ly(j))$$

$$B2b\{i,j\} = B2b\{i,j\} + B2(lx(i),lz(j))$$

Step 6: Apply DNA complement for B1', B2' to increase the complexity.

Step 7: Decode the blocks using DNA decoding for blocks B1', B2'.

Step 9: Combine the blocks to generate the fixed size key for encryption.

Step 10: Encrypt the image using encryption algorithm with the generated key.

For the hyper chaotic system, beta chaotic map is considered. Some of the characteristics of beta chaotic maps are,

- Highly sensitive to its parameters
- Strong chaotic behavior
- More efficient and resistant to several attacks

4.1 Scrambling

Multimodal input images are taken for test cases where face and hand are considered for key generation. The scrambling output of input images are shown in Fig.2.

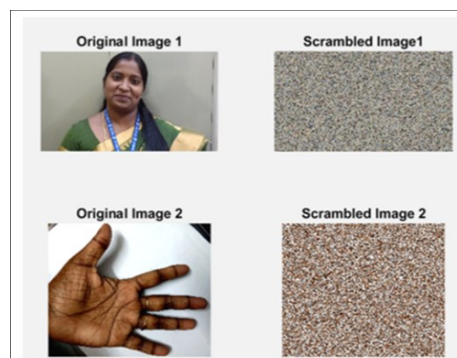


Fig 2 – Original and scrambled images of multimodal biometrics

4.2 Code generated using DNA Sequences

The four nucleic acid bases of DNA sequences are A, C, T, G; where A stands for Adenine, C for Cytosine, G for Guanine, T for Thymine. Here, A is complementary to T and G is complementary to C. There are about 24 types of coding schemes in nucleic acid bases in which Watsonrick complementary rule consists of 8 coding schemes as shown in table 3. Complementary rule is classified in to 6 groups as shown below. DNA sequences carry information using biological techniques to achieve encryption. [7, 17, 19, 28, 29, 33]

Table 3 – DNA COMPLEMENT

(AC)(TC)(CG)(GA)	(AT)(TG)(GC)(CA)
(AC)(CT)(TG)(GA)	(AC)(CG)(GT)(TA)
(AG)(GT)(TC)(CA)	(AG)(GC)(CT)(TA)

Now the generated key can be used for encryption. Encoded output after DNA decoding [22, 26, 27] is shown in Fig 4. After applying SHA, 256 bit key is generated and is shown below.

ACGATGTGGGTGGGGGGA GGA CGA GCACCA GGGGCGGGT TCCCGGGTCCGCA GA GA GGTGCGGGGCGGGC
AGGGCCCA GCA CGCGCCGCGA GA CGA GGA GCGGATTGTGA TGGGGGA GGGGGCTGGGGCGGGCCCA CCGG
TGGACCA GCGTA GAA GCGA GGCGGGTGTGTGGCA GTGCGGA A GGGGTCCGAATCCA A GCA GGGGTGA GGCC
AGAAGCCCGCGTCGCGCGCGGGACTCCGCA GCACTGGGCGACGAA

Above is the key generated in DNA sequence which are represented using DNA variable.

0011011000110110000100100110010001110000001001001100010011010100110010000100100110001001101
010011001100010010011000100110011000000010010011001000110011001100010001001001000101010011
01010001001001100100011010100110101000100100110010001101010011010100010011000100110001001100
010010011000100110010001100100001001001100100011010100110101000100100110010001101000011001100010010
01100100011010

Above is the binary version of 256 bits key generated. The output of the proposed system in shown in Fig.4

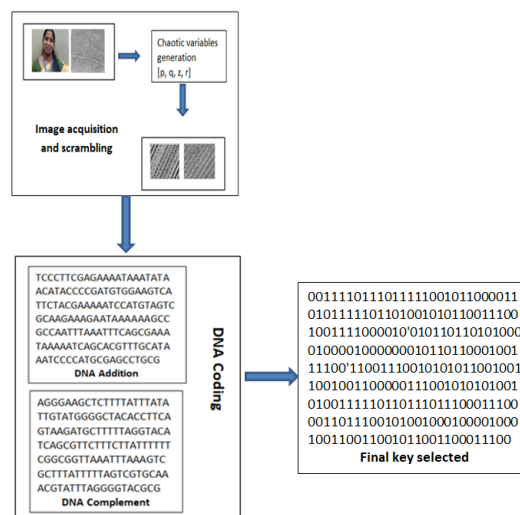


Fig 4 – Final accepted key

4.3 Security Analysis

Security analysis is used to find the strength of the key generated. Analysis of security techniques are mainly based on cryptosystem. Few security attacks [20, 23, 24, 30] on encrypted image are discussed below:

4.3.1 Key Space Analysis

When the key size is less, based on the number of trials there are possibility for the key to be identified. Hence the key size should be large which shouldn't be guessed or identified by any attacks. In our proposed system, key size is 256 bits which is resistant to Brute force attack.

4.3.2 Correlation co-efficient Analysis

Correlation coefficient is calculated between an image and the same image processed using standard deviation filtering. Correlation co-efficient calculated as in equation(6). Values obtained proves that the generated key is resistant to statistical attacks.

$$\begin{aligned}
 E &= \frac{1}{N} \sum_{i=1}^N x_i \\
 D(x) &= \frac{1}{N} \sum_{i=1}^N (x_i - E(x))^2 \\
 Cov(x, y) &= \frac{1}{N} \sum_{i=1}^N (x_i - E(x)) \sum_{i=1}^N (y_i - E(y)) \\
 r_{xy} &= \frac{cov(x,y)}{\sqrt{D(x)}\sqrt{D(y)}}
 \end{aligned} \tag{6}$$

4.3.3 Information entropy Analysis

Entropy measures the complexity of the system and is calculated as shown in equation(7). If the entropy value of the encrypted image is approximately 8, then the system is more complex. Here, in the proposed system the scrambled image entropy value proves that the proposed system is complex.

$$H(x) = - \sum_{i=1}^n (x_i \log p(x_i)) \tag{7}$$

Where, x denotes set of symbols, n denotes the total number of symbols, x_i denotes the probability of x_i in x.

4.3.4 Differential Analysis

Differential analysis calculates change of pixels in the encrypted image. Number of pixels change rate [NPCR] is used for calculating the change of pixels in encrypted image to its original image. Unified average change intensity [UACI] calculates average change of pixels. If the values of these parameters are higher, then there will be more variations in the encrypted image for small variation in original image. NPCR and UACI are calculated as:

$$NPCR = \frac{\sum_{i=0}^{M-1} \sum_{j=0}^{N-1}}{M \times N} \times 100\% \tag{8}$$

Where, D is the auxiliary matrix created when $C1(i, j) = C2(i, j)$, then $D(i, j) = 0$, otherwise $D(i, j) = 1$.

4.3.5 Results

Correlation co-efficient analysis and Entropy value are given in table 4 and table 5 for test inputs from the dataset. Security analysis of various inputs are compared in table-6. The comparison of inputs taken in references 1 – 5 are compared with proposed system input values. in table 7.

Table 4 - Correlation value

Input	Correlation value
Input image 1	0.0029
Input image 2	0.0114

Table 5 - Entropy value

Input	Encrypted Image Entropy value
Input image 1	7.2364
Input image 2	7.2968

Table 6 – Security analysis of various inputs

IMAGE	CORRELATION	ENTROPY	PSNR	MSE	NPCR	UACI
Airplane	0.002917122	6.663908	12.26333205	3861.421018	0.984704	0.183782387
Boat	0.007154649	7.19137	11.76959939	4326.337753	0.990414	0.194394789
Cameraman	0.020213095	7.009716	9.313678662	7615.746765	0.988678	0.264166067
House	-0.006930888	7.485787	10.30725426	6058.358032	0.992835	0.23846653
Lena	-0.004436002	7.779525	9.760363543	6871.380634	0.995137	0.263148176
Pepper	0.003627721	7.740886	8.894229604	8387.980273	0.994752	0.29494436
Baboon	-0.009721656	7.702034	10.66386753	5580.763365	0.994489	0.236170328
Fruit	0.00637657	7.91232	7.76274389	10884.43705	0.995422	0.331558038

Table 7 – Entropy and Correlation value comparison

IMAGE	REFERENCE	ENTROPY VALUE	CORRELATION
Pepper	Ref. 1	7.9993	0.265
Lena	Ref. 2	7.9993	0.0214
Lena	Ref. 3	7.9994	0.0015
Lena	Ref. 5	7.9982	-0.0012
Lena	Proposed system	7.7795	-0.0044

5. Conclusion

Biometric template is used as the reference for generating 256 bit key which is used for Image encryption. Chaotic maps and DNA sequences are used for implementation which is highly secure. Several cryptanalysis are done to prove the key is very secure and sensitive to initial conditions. The generated key can be used for biometric encryption and calculate FAR, FRR for extension of the work. The fusion combinations for the modalities are evaluated with less False Acceptance Rate (FAR), False Rejection Rate (FRR) and Equal Error Rate (EER). The main purpose of the proposed system is to reduce the EER as low as possible and improve the performance of the system by achieving error free decryption.

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REFERENCES

- [1] Y. Zhang, "Cryptanalysis of a novel image fusion encryption algorithm based on dna sequence operation and hyper-chaotic system," *Optik – International Journal for Light and Electron Optics*, vol. 126, no. 2, pp. 223–229, 2015.
- [2] P. Zhen, G. Zhao, L. Min, and X. Jin, "Chaos-based image encryption scheme combining dna coding and entropy," *Multimedia Tools and Applications*, vol. 75, no. 11, pp. 6303–6319, 2016.
- [3] L. zeng and R. Liu, "Cryptanalyzing a novel couple images encryption algorithm based on {DNA} subsequence operation and chaotic system," *Optik - International Journal for Light and Electron Optics*, vol. 126, no. 24, pp. 5022 – 5025, 2015.
- [4] Rim Zahmoul, Ridha Ejbal, Mourad Zaied, Image encryption based on new Beta chaotic maps, *Optics and Lasers in Engineering*, Volume 96, 2017, Pages 39-49, ISSN 0143-8166
- [5] Lifang Wu, Xing Sheng Liu, Songlong Yuan, Peng Xiao, "A Novel Key Generation Cryptosystem Based on Face Features", *International Conference on Signal Processing*, pp. 1675 -1678, 2010.
- [6] Arunjain, Sonaarrarwal, " Multimodal biometric system: A survey", *International journal of applied science and advance technology*, pp 58 – 63, vol no. 1, issue no. 1, June 2012.
- [7] Christian Rathgeb, Andreas Uhl, "A survey on biometric cryptosystems and cancelable biometrics", *EURASIP Journal on information security*, pp 1 – 25, 2011. <http://jis.eurasipjournals.com/content/2011/1/3>.
- [8] Juan ramon troncso-pastoriza, Daniel Gonzalez-jimenez, Fernando perez-gonzalez, "Fully private non-interactive face verification", *IEEE transactions on Information forensics and security*, pp 1101 – 1114, vol no 8, issue no 7, 2013.
- [9] Xinmiao zhang, Richard shi, Jame ritcey, "On the implementation of modified fuzzy vault for biometric encryption", *IEEE Information theory and applications workshop*, pp 56 -61, 2012.
- [10] Gaurav Bhatnagar, Q. M. Jonathan Wu, "Chaos-Based Security Solution for Fingerprint Data During Communication and Transmission", *IEEE transaction on instrumentation and measurement*, pp. 876 -887, Vol. 61, Issue no. 4, 2012.
- [11] S. Sowkarthika N. Radha, "Securing Iris and Fingerprint Templates Using Fuzzy Vault and Symmetric Algorithm", *International Conference on Intelligence Systems and Computing*, pp. 189 – 193, 2013.
- [12] Julien Bringer, Hervé Chabanne, and Alain Patey, "Privacy-Preserving Biometric Identification Using Secure Multiparty Computation", *IEEE signal processing magazine*, pp. 42 – 52 Vol.30, Issue no.2, 2013.
- [13] R. Guesmi, M. A. B. Farah, A. Kachouri, and M. Samet, "A novel chaos based image encryption using dna sequence operation and secure hash algorithm sha-2," *Nonlinear Dynamics*, vol. 83, no. 3, pp. 1123–1136, 2016.
- [14] P. Zhen, G. Zhao, L. Min, and X. Jin, "Chaos-based image encryption scheme combining dna coding and entropy," *Multimedia Tools and Applications*, vol. 75, no. 11, pp. 6303–6319, 2016.
- [15] A. Jain and N. Rajpal, "A robust image encryption algorithm resistant to attacks using dna and chaotic logistic maps," *Multimedia Tools and Applications*, vol. 75, no. 10, pp. 5455–5472, 2016.
- [16] M. R. Saranya, A. K. Mohan, and K. Anusudha, "Algorithm for enhanced image security using dna and genetic algorithm," in *Signal Processing, Informatics, Communication and Energy Systems (SPICES)*, 2015 IEEE International Conference on, pp. 1–5, 2015.
- [17] J. Peng, S. Jin, L. Lei, and Q. Han, "Research on a novel image encryption algorithm based on the hybrid of chaotic maps and dna encoding," in *Cognitive Informatics Cognitive Computing (ICCI*CC)*, 2013 12th IEEE International Conference on, pp. 403–408, 2013.
- [18] Y. Zhou, L. Bao, and C. P. Chen, "A new 1d chaotic system for image encryption," *Signal processing*, vol. 97, pp. 172–182, 2014.
- [19] X. Jin, Y. Tian, C. Song, G. Wei, X. Li, G. Zhao, and H. Wang, "An invertible and anti-chosen plaintext attack image encryption method based on dna encoding and chaotic mapping," in *Chinese Automation Congress (CAC)*, 2015, pp. 1159–1164, 2015.
- [20] Y. Zhang, "Cryptanalysis of a novel image fusion encryption algorithm based on dna sequence operation and hyper-chaotic system," *Optik – International Journal for Light and Electron Optics*, vol. 126, no. 2, pp. 223–229, 2015.
- [21] Q. Zhang, L. Guo, and X. Wei, "A novel image fusion encryption algorithm based on {DNA} sequence operation and hyper-chaotic system," *Optik – International Journal for Light and Electron Optics*, vol. 124, no. 18, pp. 3596 – 3600, 2013.
- [22] "Image encryption using {DNA} addition combining with chaotic maps," *Mathematical and Computer Modelling*, vol. 52, no. 1112, pp. 2028 – 2035, 2010.
- [23] X. Wang and L. Liu, "Cryptanalysis of a parallel sub-image encryption method with high-dimensional chaos," *Nonlinear Dynamics*, vol. 73, no. 1, pp. 795–800, 2013.
- [24] F. zkaynak, A. B. zer, and S. Yavuz, "Security analysis of an image encryption algorithm based on chaos and dna encoding," in *Signal Processing and Communications Applications Conference (SIU)*, 2013 21st, pp. 1–4, April 2013.
- [25] B. Norouzi, S.M. Seyedzadeh, S.Mirzakuchaki, and M. R. Mosavi, "A novel image encryption based on hash function with only two-round diffusion process," *Multimedia Systems*, vol. 20, no. 1, pp. 45–64, 2014.
- [26] L. Liu, Q. Zhang, and X. Wei, "An RGB image encryption algorithm based on DNA encoding and chaos map," *Computers & Electrical Engineering*, vol. 38, no. 5, pp. 1240–1248, 2012.
- [27] H. Hermassi, A. Belazi, R. Rhouma, and S. M. Belghith, "Security analysis of an image encryption algorithm based on a dna addition combining with chaotic maps," *Multimedia Tools and Applications*, vol. 72, no. 3, pp. 2211–2224, 2014.

- [28] “A fast color image encryption algorithm based on coupled two-dimensional piecewise chaotic map,” *Signal Processing*, vol. 92, no. 5, pp. 1202 – 1215, 2012.
- [29] “A {RGB} image encryption algorithm based on total plain image characteristics and chaos,” *Signal Processing*, vol. 109, pp. 119 – 131, 2015.
- [30] “Cryptanalyzing a novel image fusion encryption algorithm based on {DNA} sequence operation and hyper-chaotic system,” *Optik – International Journal for Light and Electron Optics*, vol. 125, no. 4, pp. 1562 – 1564, 2014.
- [31] D.Jagadiswary, “Authentication Using Fused Multimodal Biometric,” *International Conference on Computational Modelling and Security (CMS 2016)*, *Procedia Computer Science*, vol. 85, no. 109 - 116, pp. 1259– 1284, 2016.
- [32] Cheng-chi lee, Chi-tung chen, Ping-hsien wu, Te-yu chen, “Three factor control protocol based on elliptic curve cryptosystem for universal serial bus mass storage devices”, *Institution of Engineering and Technology*, pp. 48 – 55, vol.7, Issue no 1, 2013.
- [33] Radhika k.R, Nalini M.K, “Biometric Image encryption using DNA sequences and chaotic systems”, *2017 International Conference on Recent Advances in Electronics and Communication Technology*, IEEE, 2017.
- [34] Shuqin Zhu and Congxu Zhu, “Secure Image Encryption Algorithm Based on Hyperchaos and Dynamic DNA Coding”, *Journal of Entropy*, 2020, 22, 772.